Concept-Based Explainable AI in Retinal Imaging: A State-of-the-Art Review of Methods, Challenges, and Future Directions

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Abstract

In high-stakes domains like healthcare, establishing trustworthy AI decisions is critical to foster confidence among experts and researchers. However, standard explanation methods often fail to meet clinical needs, offering raw data, such as saliency maps, rather than human-readable text and auditable, transparent reasoning. To bridge this gap, researchers are increasingly adopting concept-based approaches that express model reasoning through human-readable concepts grounded in expert-backed clinical knowledge. Although this field has advanced rapidly in recent years, it still lacks a shared classification system and standard testing methods, particularly in ophthalmology. This gap underscores the need for collective effort to empower researchers to shape future standards. In this paper, we examine how to make these systems more reliable and propose a new framework that enhances clinical reliability by systematically auditing concept-level reasoning and aligning model outputs with verified medical knowledge. By introducing structured evaluation and accountability mechanisms, it supports safer deployment of AI systems in real-world ophthalmic diagnostics.

Keywords: Concept-Based Explainable AI, Retinal Imaging, Ophthalmology, Medical Image Analysis, Trustworthy AI, Explainable Artificial Intelligence, Concept Bottleneck Models, SMILE

1 Introduction

Deep Learning (DL) has become well established in healthcare, with models achieving, and at times exceeding, expert-level performance in disease diagnosis, including retinal diseases from fundus images [1]. This capability is critically important, as early diagnosis could prevent up to half of the 2.2 billion cases of vision impairment worldwide [2]. However, the "black box" nature of these sophisticated models creates a profound barrier to clinical translation [3]. For an ophthalmologist, a diagnostic output without a transparent and verifiable reasoning process is clinically untenable. This opacity is a fundamental obstacle to building the trust required for real-world integration into critical patient care workflows. The field of Explainable AI (XAI) has emerged as a key approach for addressing this opacity. Seminal methods like LIME, SHAP, and Grad-CAM have focused on attributing predictions to input features, typically by generating saliency maps [4–6]. The broader XAI landscape, including post-hoc, intrinsic, and hybrid interpretability paradigms, has been extensively surveyed in recent years [7]. However, concept-based explainability in medical imaging, particularly retinal image analysis, remains comparatively underexplored.

In response to these limitations, Concept-Based XAI (C-XAI) offers a paradigm shift, aligning model explanations with high-level, human-understandable concepts [8]. Instead of pointing to pixels, the C-XAI methods explain a diagnosis in terms of the clinical biomarkers an ophthalmologist would use, such as 'hard exudates', ' retinal haemorrhages', or 'optic disc pallor'. This approach not only provides more intuitive and actionable insights for domain experts but also fosters the development of inherently more robust models. Recent studies have begun to validate that concept-based models can offer improved stability and resilience against adversarial perturbations compared to their end-to-end counterparts [9]. C-XAI provides a straightforward method for evaluating a model's reasoning against accepted medical wisdom.

The C-XAI landscape remains fragmented despite this progress. While numerous surveys have mapped the broader XAI field [10, 11], and some specifically address medical XAI [12], they often lack a specific focus on concept-based methods in the medical domain, specifically retinal images. More recent works have begun benchmarking specific C-XAI techniques [13] or exploring their use in natural language [14]. However, a systematic, unified review of C-XAI methods tailored explicitly to the unique challenges of medical imaging and, in particular, retinal analysis is conspicuously absent. Although Hou et al. [12] recently surveyed Self-Explainable AI for general medical imaging, they do not address the specific, fine-grained challenges of retinal pathologies. This paper addresses this critical gap by presenting a systematic review of Concept-based XAI for retinal image analysis, synthesizing the literature from 2017 to 2026 to establish a coherent taxonomy and a forward-looking research agenda. Recent developments, such as neurosymbolic frameworks that integrate deep visual representations with structured clinical reasoning for retinal diagnosis, further highlight the growing shift toward semantically grounded explainability in ophthalmology [15]

1.1 Contributions

This work makes three pivotal contributions to the advancement of trustworthy AI in medical imaging, focusing specifically on retinal analysis while providing a framework applicable to broader clinical domains:

Fig. 1 Conceptual word cloud summarizing the central themes and methodological foundations of retinal Concept-Based Explainable AI

- **Refined Taxonomy:** This paper introduces a comprehensive taxonomy for Concept-based XAI (C-XAI) in retinal imaging that categorizes methods across three dimensions: integration stage, semantic concept type (pathological, anatomical, or rational), and output presentation. While designed for ophthalmology, this structured lens serves as a unified language to evaluate the C-XAI landscape in any medical imaging context.
- **SMILE-based Reliability for Self-Explainable C-XAI:** To prevent models from relying on non-clinical "shortcuts," this paper also proposes a framework for active auditing via SMILE, a model-agnostic tool. This layer of scrutiny ensures that self-explainable models are held accountable to clinical logic, a necessity for reliable AI in any high-stakes healthcare environment.
- **Clinically Aware Ophthalmic Decision Support Framework:** Furthermore, this paper proposes a forward-looking road-map for next-generation diagnostic systems. This framework integrates Relational Knowledge Graphs to capture structural dependencies, a Report Generator for clinical communication, and an Auditing Layer for rigorous medical verification.

1.2 Paper Organization

The subsequent parts of this work are structured as follows: Section 2 establishes the theoretical background, presenting the refined taxonomy and a review of C-XAI methods, including Post-Hoc and Explainable-by-Design models. Section 3 analyzes critical deployment challenges and evaluation metrics, while Section 4 details the proposed SMILE-based auditing methodology. The paper concludes with an overview of available resources and future research directions.

1.3 Methodology

1.3.1 Search Strategy Documentation

To ensure a comprehensive and reproducible "paper trail," the literature search for this review was conducted using a multi-phase approach across major technical and medical databases.

Databases:

PubMed was used to identify clinically oriented studies and peer-reviewed medical journals, while IEEE Xplore was used to capture engineering- and hardware-centric innovations in medical imaging. To ensure comprehensive coverage of cross-disciplinary citations, Google Scholar was included in the search strategy. Finally, arXiv was leveraged to incorporate high-impact, state-of-the-art preprints from 2024 to 2026, which define the most recent shifts in the field, such as Relational Concept Discovery and Generative Auditing.

Search Queries:

The search technique utilised a focused string intended to merge the methodological innovations of Concept-Based Explainable AI (C-XAI) with the clinical nuances of retinal imaging.

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("Concept-Based XAI" OR "Concept Bottleneck" OR "C-XAI" OR "Post-Hoc
Concept" OR "Self-Explainable AI") AND ("Retina" OR "Ophthalmology" OR
"Fundus" OR "OCT" OR "Diabetic Retinopathy")
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This particular setup was designed to encompass a wide yet pertinent array of techniques, including post-hoc Concept Activation Vectors (TCAV), intrinsic designs such as Concept Bottleneck Models (CBM), and Prototypical Part Networks. The search term included vocabulary covering multiple retinal conditions and modalities, allowing the collection of research focused on identifying clinically significant symptoms, including hard exudates, macular edema, and microaneurysms. Dataset resources such as FGADR are particularly important for retinal C-XAI because they provide fine-grained lesion annotations that support concept-level evaluation, including lesion segmentation, grading, and transferability analysis [16].

Data Range:

The systematic literature search (2017–2026) was designed to capture the whole evolution of retinal C-XAI, moving from early feature attribution to modern structural reasoning. With the emergence of fundamental techniques like SHAP in 2017, the review documents the peak of supervised Concept Bottleneck Models (CBMs) in 2022 and the subsequent "2025 Shift" toward Generative Auditing and Relational Concept Discovery.

1.4 Inclusion and Exclusion (I/E) Criteria

The investigation aimed to document the technical progression from first-feature-attribution methods, such as SHAP, to fundamental Concept Bottleneck Models (CBM) and the subsequent "2025 Shift" towards Relational Concept Discovery and Generative Auditing. Study eligibility was strictly governed by an Inclusion and Exclusion (I/E) framework that prioritized research explicitly utilizing concept-based methodologies over standard saliency maps to address ophthalmic pathologies. By focusing on novel architectures, standardized metrics, and expert human-in-the-loop evaluations, this methodology provides a clinically grounded synthesis of research targeting significant symptoms such as hard exudates, macular edema, and microaneurysms.

The article selection process was carried out in accordance with PRISMA 2020 standards. From the initial 200 records, after removing duplicates and screening based on title and abstract, 46 studies that exactly matched the C-XAI criteria in Retina were finally selected for the final analysis.

2 Background

2.1 Definition

To ensure an understanding of Concept-based Explainable AI (C-XAI), this section breaks down the core components of the pipeline shown in Figure 2. The process

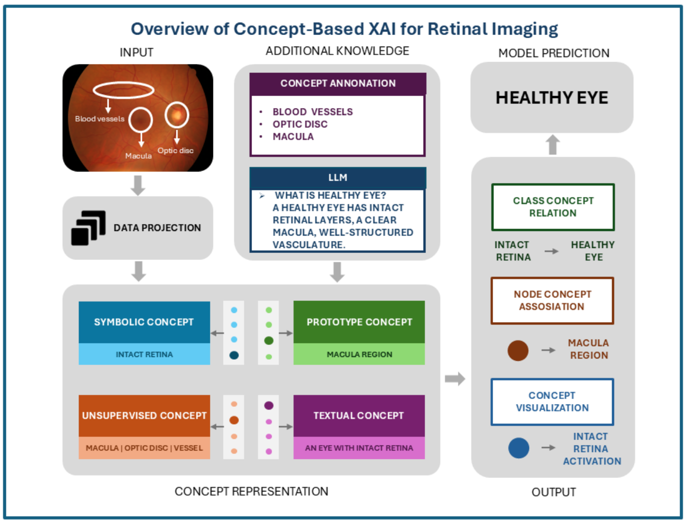

Fig. 2 End-to-end C-XAI pipeline for retinal imaging. The framework maps raw fundus image data into human-understandable clinical concepts (e.g., Macula, Optic Disc) to provide transparent diagnostic reasoning before the final prediction.

starts simply with an input fundus image. However, instead of just processing raw pixels, C-XAI models first map this data into a meaningful latent space through a Data Projection stage. The real innovation happens in the next step: the Concept Representation layer. This is where the model's internal logic is translated into ideas that actually make sense to a human.

Figure 3 summarises the main concept and explanation types used in concept-based explainability. These concepts take many forms, generally falling into five main categories.First is *Symbolic Concepts* [9], which are clear, high-level attributes, such as the presence of an *intact retina*. Then there are *Unsupervised Concepts* [17], latent patterns the model discovers on its own, often matching anatomical structures like the macula or optic disc. Furthermore, there are *Prototype Concepts* [18], which use standard examples, such as a textbook macula region, as reference points. *Textual Concepts* include rich linguistic descriptions [19], such as an eye with an intact retina. Finally, there are *Relational Concepts*, which define the structural layout, such as how scattered lesions are spatially bound together [20].

Building these concepts often requires bringing in additional knowledge to guide the model. There are several methods for this including **Concept Annotation** [8], where clinical experts manually label features such as blood vessels, or **knowledge from Large Language Models (LLMs)** that taps into their understanding to define complex states [19] such as a healthy eye.

Once the model learns these concepts, it generates output explanations to make its reasoning transparent before arriving at a final model prediction. These explanations take three forms: **Class-Concept Relation**, which tells how important a specific concept was to the diagnosis; **Node-Concept Association**, which links specific internal neural units to the human concepts they detect; and **Concept Visualisation**, which spatially grounds the findings by using heatmaps to show precisely where a concept appears on the image.

While this pipeline provides a valuable framework, it operates within a field where the precise definitions of terms such as "concept" and "explanation" remain inconsistently defined across the literature, as highlighted in foundational XAI surveys [10] [11]. This ambiguity complicates the evaluation and comparison of different methods [21], with different research communities often adopting varying interpretations of these critical terms [7].

2.1.1 What are clinical ophthalmic concepts?

The term "concept" has been defined in many ways in the concept-based explainability literature. As noted by Molnar [22], "a concept can be any abstraction, such as a color, an object, or even an idea." Aligning with this broad understanding and further elaborating upon the typologies discussed later, this paper treats ophthalmic concepts as clinically meaningful visual or semantic units that can support diagnosis, explanation, and model auditing in retinal imaging.

In retinal image analysis, these concepts commonly include anatomical structures, pathological biomarkers, and image-level clinical signs. Anatomical concepts include the optic disc, macula, fovea, and retinal blood vessels. Pathological concepts include microaneurysms, haemorrhages, hard exudates, cotton-wool spots, drusen, neovascularisation, and optic disc pallor. These concepts are important because they correspond

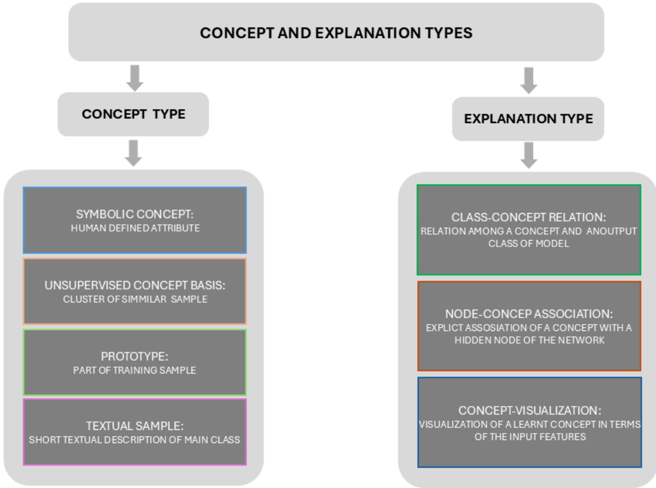

Fig. 3 Concept and explanation types used in concept-based explainability (C-XAI) methods.

to the visual evidence that ophthalmologists use when assessing conditions such as diabetic retinopathy, glaucoma, and age-related macular degeneration.

The clinical value of these concepts is not only descriptive but also diagnostic. For example, the presence and distribution of microaneurysms and haemorrhages may support diabetic retinopathy grading, while optic disc changes can be relevant for glaucoma assessment. Similarly, drusen and macular abnormalities may indicate age-related macular degeneration. Therefore, concept-based explanations are useful only when they reflect clinically meaningful evidence rather than irrelevant image artefacts or spurious correlations.

This is particularly important in high-stakes ophthalmology, where an incorrect or poorly explained diagnosis can delay treatment, increase unnecessary referrals, or contribute to avoidable vision loss. A trustworthy retinal AI system should therefore not only predict a disease label, but also provide explanations that clinicians can inspect, question, and compare with accepted ophthalmic knowledge.

2.1.2 What is a concept-based explanation in ophthalmic diagnosis?

Even when considering a fixed concept, the definition of a concept-based explanation can be elusive. However, existing literature largely concurs that concept-based explanations should illuminate how Deep Neural Networks arrive at decisions by utilizing concepts [23]. In ophthalmic diagnosis, this means that the explanation should connect the model's prediction to clinically meaningful retinal evidence, rather than only highlighting pixels or low-level visual patterns.

For example, a concept-based explanation should indicate whether a diagnosis is supported by relevant findings such as microaneurysms, haemorrhages, hard exudates, vessel abnormalities, macular changes, or optic disc features. This is different from a generic saliency map, which may show visually active regions without explaining whether those regions correspond to clinically meaningful signs. In this sense, concept-based explanations aim to make the model's reasoning more understandable and clinically inspectable.

A useful ophthalmic explanation should therefore be meaningful, coherent, explicit, and relevant to the diagnostic task [17, 24]. It should help clinicians assess whether the model is relying on appropriate medical evidence or on misleading shortcuts, such as image artefacts, blur, or dataset-specific biases. This is especially important in retinal imaging, where small pathological signs can have major implications for diagnosis, referral, and treatment decisions.

2.2 Methodological Approaches: Post-Hoc vs. Explainable-by-Design C-XAI in Retina

Post-Hoc Concept-Based Explanation Methods:

Post-hoc C-XAI methods are investigative tools that are applied to a pre-trained, and often opaque, deep neural network to audit its decisions using a human-understandable vocabulary. The core principle is to analyze the model's behavior without altering its architecture. This is typically achieved by taking a set of inputs that represent a specific clinical concept. The main advantage of this method is flexibility; it allows researchers and clinicians to examine any existing model, offering a powerful way to diagnose failures or build confidence in a system that is already deployed. However, its main limitation is that the explanation is an approximation, a retrofitted interpretation that may not perfectly represent the model's true, and potentially more complex, internal reasoning [25].

Explainable-by-Design Concept-Based Models:

Explainable-by-design C-XAI models are architecturally engineered from the ground up for transparency. These models do not treat the explanation as a separate analysis but as an integral, causal step in their decision-making process. A typical design involves an intermediate "concept layer" within the neural network that is explicitly trained to detect specific clinical concepts. The explanation is therefore not an interpretation of the model's decision, but a direct readout of its reasoning. This provides a very high degree of faithfulness, but at the cost of flexibility, as it requires building a new, specialized model and cannot be used to audit existing systems [26]. While explainable-by-design models are theoretically robust, empirical evaluations in retinal tasks [32] reveal that their performance is susceptible to the granularity and quality of concept labels, often facing a steeper accuracy trade-off compared to post-hoc methods in heterogeneous clinical datasets. Table 1 summarises the main methodological and clinical differences between post-hoc and explainable-by-design C-XAI approaches.

2.3 Taxonomy of Concept-based Explainability Approaches

In this section, a detailed taxonomy of concept-based explainable AI (C-XAI) methods for retinal images is presented as illustrated in Figure 4, designed to guide practitioners

Table 1 Comparison of Post-Hoc and Explainable-by-Design Concept-Based XAI approaches across key methodological and clinical dimensions.

Category	**Post-Hoc C-XAI**	**Explainable-by-Design C-XAI**
Architectural Dependency	Model-Agnostic [27]	Model-Specific [8, 9]
Methodological Paradigm	Observational [4]	Interventional [19]
Explanatory Faithfulness	Aspirational and Unverified [28]	High and Inherent [29]
Robustness of Explanation	Vulnerable [30]	Robust [26]
Clinical Application Focus	Debugging [25]	Certification [29]
Exemplar Methods	TCAV [27] / ACE [17] / Post-hoc concept discovery [31]	CBMs [8] / Label-Free CBMs [19] / Logic-Based CBMs [9]

through the nuanced terrain of C-XAI. Actionable decision criteria grounded in development constraints and ethical considerations are also presented that help in selecting the method that best balances technical feasibility with the clarity and trustworthiness of the explanations. To provide a systematic and thorough evaluation, each concept-based explainability method and model is characterised using a comprehensive set of analysis dimensions, explicitly applied to their implementation in key retinal datasets.

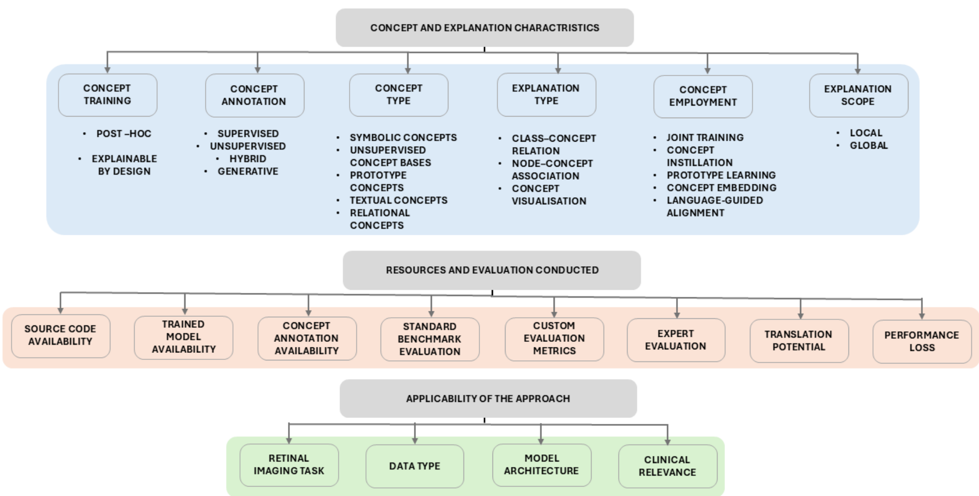

Fig. 4 Overview of the proposed taxonomy of Concept-Based Explainability (C-XAI) approaches for retinal imaging. The taxonomy organises methods according to concept and explanation characteristics, applicability to retinal tasks, and resources/evaluation dimensions, providing a structured basis for comparing C-XAI methods across methodological, clinical, and reproducibility criteria.

2.3.1 Concept and Explanation Characteristics

To systematically evaluate the C-XAI landscape, we first characterize approaches according to their integration stage, concept type, annotation strategy, explanation type, explanatory scope, and concept employment mechanism. These dimensions provide a structured way to compare concept-based methods across different retinal imaging tasks and model architectures.

The dimension of **Concept Training** distinguishes whether concepts are applied post-hoc to interpret an already trained model, or explicitly integrated during training within an explainable-by-design architecture. Post-hoc methods use concepts after model training to probe or audit the behaviour of an existing black-box model, whereas explainable-by-design approaches introduce concepts as part of the model architecture or training process.

Methods are further differentiated by their**Concept Annotation** source.Supervised approaches rely on expert-annotated concept sets, such as blood vessels, optic disc, macula, microaneurysms, haemorrhages, hard exudates, or drusen. Unsupervised approaches discover latent structures without explicit labels, often revealing patterns that may correspond to anatomical regions or pathological signs. Hybrid approaches combine limited supervised annotations with unsupervised discovery, while generative approaches automatically produce candidate concepts using foundation models or large language models [33, 34].

Across these strategies, concepts generally appear in five principal typologies. **Symbolic Concepts** refer to explicitly defined clinical attributes with human-understandable meanings, such as retinal haemorrhages, hard exudates, optic disc pallor, or macular abnormalities. **Unsupervised Concept Bases** capture emergent latent patterns that often correspond to anatomical structures, such as vessel clusters, macular regions, or optic-disc-related patterns. **Prototype Concepts** represent archetypal examples learned directly from the data, for instance a prototypical presentation of severe cupping in glaucoma or a typical cotton-wool spot. **Textual Concepts** encode linguistic clinical descriptions into embedding representations, such as "bright optic disc", "clear macula", or "normal blood vessels". Finally, **Relational Concepts** model structural dependencies and spatial binding among distributed retinal features, such as how scattered micro-lesions are spatially arranged in diabetic retinopathy [20].

Regarding the output, concept-based explanations primarily take three forms. A **class–concept relationship** quantifies how strongly a concept contributes to a final diagnostic prediction. For example, a method may estimate how much the presence of haemorrhages or abnormal vessels influences the classification of diabetic retinopathy. A **Node-Concept Association** maps internal neural units, filters, or hidden representations to human-understandable concepts, helping to identify which parts of the network respond to retinal structures or pathological signs. A **Concept Visualisation** provides spatial grounding by showing where a concept appears in the image, often through heatmaps, activation maps, prototype patches, or concept-specific visual evidence.

A useful concept-based explanation should not only expose the presence of concepts, but also clarify how these concepts support the model's decision. Therefore, explanation quality depends on whether the concept is meaningful, coherent, clinically relevant, and explicit. In retinal imaging, this is particularly important because an explanation must be understandable to clinicians and should align with ophthalmic reasoning rather than simply highlighting visually salient pixels.

Explanation scope refers to how broadly a model's reasoning is explained.**Local explanations** explain why the model made a decision for a single case, such as identifying lesions or abnormal vessels in one fundus image.**Global explanations** describe general patterns in the model's behaviour across many cases, for example showing how important blood vessels, optic disc changes, or macular abnormalities are for a diagnostic class overall.

For explainable-by-design models specifically, the mechanism of **Concept Employment** is also analysed, checking whether concepts are learned via joint training, concept instillation, prototype learning, concept embedding, or language-guided alignment [35]. Finally, **Performance Loss** [36] is assessed to determine whether an interpretable architecture compromises predictive accuracy relative to a traditional black-box model. While this trade-off has been a significant barrier, recent architectural innovations such as Concept Embedding Models (CEMs) have demonstrated that it is possible to achieve high interpretability without sacrificing state-of-the-art diagnostic performance [26].

2.3.2 Applicability of the Approach

From a practical standpoint, these dimensions clarify exactly which retinal data modalities the method can handle, the clinical or diagnostic tasks it is optimized for, and the architectural choices, such as specific CNN backbones or transformer models that underpin its operation.

- Data Type [37]: This specifies the data formats the method can handle. For retinal imaging, this predominantly includes images but also extends to text, graphs, tabular data, videos, and time series in broader contexts.
- Task [8]: This identifies the main computational task the method is designed for, primarily classification in retinal diagnostics, but also potentially regression or clustering.
- Network Type [20]: This refers to the specific neural network architecture employed, including CNNs for image processing, 3D-CNNs, autoencoders, Graph Neural Networks (GNNs), Multi-Layer Perceptrons (MLPs), and Transformers.

2.3.3 Resources and Evaluation Conducted

Finally, to assess the reproducibility and translational potential of each method, the availability of supporting resources and the rigour of the evaluation framework are also analysed. This dimension considers whether the authors provide source code, trained models, retinal datasets, concept annotations, or sufficient implementation details to allow the method to be reproduced and compared. Studies are also categorised according to their evaluation strategy, distinguishing between those that use standard benchmarks [38] and those that introduce new metrics for qualities such as faithfulness, robustness, concept purity, or explanation stability. Crucially, this review also considers whether domain specialists have conducted a human evaluation of the method. In high-stakes ophthalmology, this dimension is vital because it shifts the assessment from theoretical performance to tangible clinical utility, trust, and expert usability.

2.4 Guidelines For Selecting A Suitable Method for Retinal Application

This section provides practical guidelines to facilitate the selection of an appropriate C-XAI method for specific research or clinical development needs within retinal image analysis. The selection process is organised around three key questions that practitioners should consider when choosing a relevant category of C-XAI methods. These questions relate to: (1) whether the learning model can be modified, (2) whether annotated clinical concepts are available, and (3) what type of explanation or concept representation is required. This selection process is summarised in Figure 5. The first

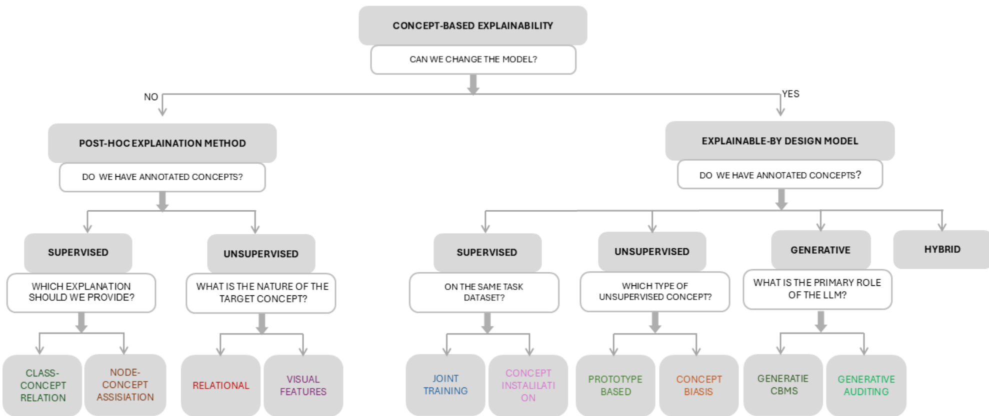

Fig. 5 Categorization Of The C-XAI Methods with Guidelines For Selecting A Suitable Approach

question is whether the model architecture can be modified or retrained from scratch. This is the initial coarse-grained decision. If the practitioner needs to work with an already trained model that cannot be modified, post-hoc concept-based explanation methods are the most suitable option. These methods allow an existing black-box model to be audited without changing its architecture. Conversely, if the model can be created or retrained from scratch, explainable-by-design concept-based models can be used, because they explicitly integrate concepts into the model architecture during training.

The second question is whether an annotated dataset of human-interpretable concepts is available for training or evaluation. If expert-annotated retinal concepts are available, such as blood vessels, optic disc, macula, microaneurysms, haemorrhages, hard exudates, or drusen, supervised concept-based approaches can be used. If only limited concept annotations are available, hybrid approaches may be more appropriate because they combine a small number of supervised concepts with unsupervised concept discovery [39]. If no concept annotations are available, generative approaches can be considered because they dynamically create candidate concepts through an external model, often a Large Language Model (LLM). In this setting, Generative Auditing frameworks [34] can also support safety certification by cross-referencing generated concepts against verified medical literature, thereby reducing the risk of hallucination.

The third question concerns the desired explanation outcome within the selected method category. Practitioners should determine whether they need class–concept relations, node–concept associations, visual concept representations, or a specific form of concept annotation. For example, if a post-hoc supervised approach is selected, methods differ according to whether they provide Class–Concept Relations or Node–Concept Associations [40]. Similarly, explainable-by-design supervised approaches differ according to whether concept annotations are required on the same dataset used for the primary diagnostic task or can be obtained from a separate dataset [41].

In the case of unsupervised methods, practitioners must also consider how emergent concepts are represented. For diffuse retinal pathologies such as diabetic retinopathy, it is important to distinguish between identifying isolated visual features and understanding their spatial connectivity. If the diagnostic goal requires capturing the structural binding of scattered lesions, Relational Concept Discovery methods should be prioritised over simpler feature-discovery approaches. Finally, if an unsupervised explainable-by-design approach is chosen, methods within this category differ primarily in how they represent the adopted unsupervised concepts [42].

2.5 Post-Hoc Concept-Based Explanation Methods

Post-hoc concept-based methods are analytical tools designed to explain the behaviour of an existing, already trained model without modifying its internal architecture. As their name implies, these methods analyse the model after training, treating it as a fixed black-box system. Their primary goal is to explain the model's behaviour by mapping its internal representations or final predictions onto a vocabulary of human-understandable concepts.

When expert-defined concepts are available, supervised post-hoc approaches use datasets with explicit concept annotations, such as labels for *blood vessels*, *optic disc*, or *macula*. The method then explains the model's prediction by quantifying the relationship between a clinical concept and the predicted class. For example, it may estimate how important the presence of an *intact retina* is for a final *healthy eye* prediction. This can also involve node–concept association, where an internal neuron is found to be highly activated by a specific concept, such as the *macula region*.

When expert labels are not available, post-hoc methods can derive explanations from an unsupervised concept basis. For instance, a method may automatically discover concepts that correspond to central anatomical regions, such as the *macula*, *optic disc*, or *retinal vessels*, by clustering the model's internal activations. It can then explain the model's prediction based on the presence and importance of these automatically discovered features. In both supervised and unsupervised settings, post-hoc methods may provide concept importance scores, node–concept associations, or concept visualisations.

- **Advantages:** Post-hoc methods are flexible because they can be applied to already trained and clinically validated models without modifying their architecture. This makes them useful for auditing existing diagnostic systems. They also preserve the predictive performance of the original black-box model while adding an additional layer of interpretability for clinical inspection.
- **Limitations:** However, post-hoc explanations should be interpreted with caution. Since these methods are applied after training, they cannot always guarantee that the identified concepts were causal in the model's decision. Instead, they may capture correlations between clinical concepts and model outputs, meaning that their faithfulness, stability, and vulnerability to misleading explanations require careful evaluation in high-stakes retinal diagnosis.

2.5.1 Supervised Concept-Based Explainability Methods

These methods analyze a neural network's behavior by examining its responses to meticulously annotated samples with symbolic concepts. The core premise behind these methods is that during training for a primary task (e.g., retinal image classification), a deep learning network may implicitly learn to recognize certain human-understandable concepts (e.g., blood vessels, optic disc, macula). These methods aim to assess which concepts the network has learned precisely.

The specific analysis conducted varies based on the type of explanation sought. Some methods analyse how the model's current prediction, or the overall weighting of an entire output class, correlates with the projection of a specific concept within the model's latent space. This helps quantify the importance of a concept to a particular diagnosis. Other methods investigate the activations of individual hidden nodes or filters within the network to establish direct associations with a given concept for extracting node-concept associations. This reveals which internal network components are responsible for detecting specific retinal features.

Class-Concept Relation Methods

, Supervised post-hoc methods, which rigorously examine the connection between the model's output classes and a predefined set of symbolic concepts, focus on class-concept relation explanations.Figure 6 illustrates how class concept relation methods quantify the influence of clinical biomarkers on retinal diagnostic outcomes using TCAV-based sensitivity analysis.This is generally achieved by presenting specific samples known to represent these concepts to the already trained network and then analyzing its behavior. In the context of retinal images, this would involve a model trained to diagnose eye conditions that would be probed with image patches representing relevant concepts (e.g., a segment of "normal blood vessels" or a "clear macula"). The process showcases an input fundus image, additional knowledge through concept annotation of blood vessels, data projection, and concept representation, ultimately leading to a class-concept relation output where 'Normal blood vessels' influence the 'Healthy eye retina' classification with a score of 0.8.

To quantify the relationship between clinical concepts and model predictions, the **Conceptual Sensitivity** framework has been applied. First, for an individual input x, the sensitivity of the class k to a specific concept C is defined as the directional derivative of the model's logit f_k in the direction of the Concept Activation Vector v_C:

$$S_{C,k}(x) = \nabla f_k(h_l(x)) \cdot v_C \tag{1}$$

where $h_l(x)$ denotes the activations of the network at layer l.

The **Testing with Concept Activation Vectors (TCAV)** score provides a global measure of how strongly a predefined human-understandable concept influences a model's prediction for a given class. It quantifies the proportion of samples for which increasing the presence of a concept leads to an increase in the model's output for that class, thereby offering an interpretable estimate of concept importance.

To derive this global importance score (e.g., the 0.8 score for 'Normal blood vessels'), we compute:

$$TCAV_{C,k} = \frac{|\{x \in X_k : S_{C,k}(x) > 0\}|}{|X_k|} \tag{2}$$

In Equation 2, X_k represents the set of all inputs belonging to class k, and the score quantifies the fraction of samples where the presence of the concept C increases the prediction probability for that class.

Despite its widespread adoption, standard TCAV [27]is often limited by the assumption of linear concept separability and susceptibility to 'directional divergence' in noisy latent spaces. To address these instabilities, Visual-TCAV [43]introduces contrastive expectations to stabilize the Concept Activation Vectors (CAVs) by accounting for visual context, which is particularly beneficial for distinguishing subtle retinal lesions from complex backgrounds. Furthermore, Concept Semantic Re-arrangement (CSR) [44]has been proposed to better capture non-linear relationships between classes and concepts by aligning semantic information through logic-guided reasoning. This enables biomarkers with high morphological variability to be represented more faithfully within the model's decision-making process.

This transition from local directional derivatives for a single instance x to the aggregate TCAV score allows the framework to bridge the gap between individual image evidence and global model transparency.

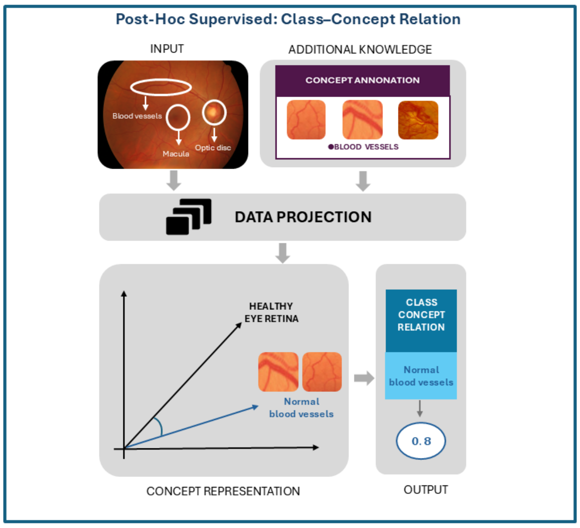

Fig. 6 Quantifying clinical concept importance via Class-Concept Relations. This post-hoc method calculates the influence of biomarkers like blood vessels on retinal diagnostic outcomes using TCAV-based sensitivity analysis.

Node-Concept Association methods

A variety of post-hoc methods have been developed to associate a model's internal units with specific, human-understandable concepts.Figure 7 Shows how node concept association methods map internal neural units to clinically meaningful retinal features. In a clinical context, this is akin to conducting a neuro scientific study on the AI model itself to identify which parts of its 'brain' have developed into specialist detectors [45] for pathologies or landmarks. This section reviews the evolution of these methods, from early manual approaches to more sophisticated, automated techniques [44].

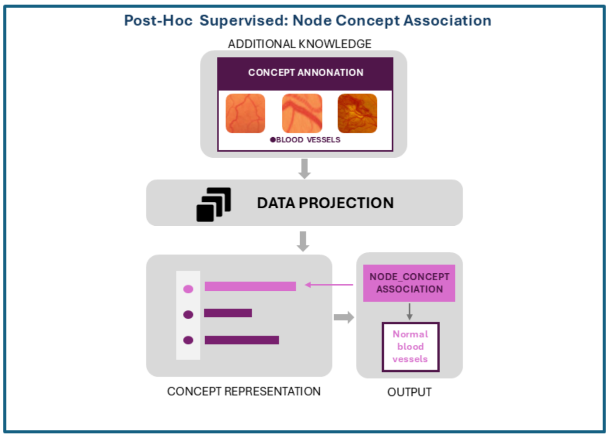

Fig. 7 Mapping clinical concepts to internal neural units.The framework clarifies the spatial association between hidden nodes and retinal features (e.g., blood vessels), revealing how the network internalizes specific ophthalmic landmarks.

Supervised post-hoc C-XAI methods examine whether a trained black-box model has encoded clinically meaningful concepts within its internal representations. These methods differ in the type of explanation they produce: some quantify class–concept relationships, others estimate causal concept effects, and others associate internal neurons with human-interpretable retinal features. Table 2 summarises these representative frameworks according to their explanation type, core mechanism, and implementation strategy.

Table 2 Summary of supervised post-hoc concept-based explainability frameworks, core mechanisms, and implementation strategies.

Framework	**Category**	**Core mechanism**	**Implementation strategy**
TCAV	Class–Concept relation	Measures global conceptual sensitivity using directional derivatives to estimate the influence of a concept on class predictions [27].	Learns Concept Activation Vectors (CAVs) through linear probes in latent space; later extensions refine contrastive sensitivity and semantic alignment [44, 46].
Visual-TCAV	Class–Concept relation	Improves the stability of concept sensitivity estimates by addressing directional divergence in noisy latent spaces [44].	Uses image-specific gradients to stabilise CAV orientation, making the method more suitable for visually noisy retinal images.
CSR	Class–Concept relation	Models non-linear class–concept relationships through semantic re-arrangement and logic-guided alignment [46].	Re-aligns concept representations so that complex clinical morphologies can be mapped more faithfully to model reasoning.
CARs	Class–Concept relation	Represents complex or multi-modal concepts by relaxing the assumption that concepts are linearly separable [47].	Uses non-linear probes, such as kernel SVMs, to define concepts as bounded regions in latent space rather than single concept vectors.
IBD	Class–Concept relation	Decomposes class predictions into weighted combinations of multiple concept directions, exposing both collective and residual influences [48].	Reconstructs class representations using a learned concept basis, producing interpretable contribution weights for each concept.
CaCE	Causal explanation	Estimates the causal effect of a concept through explicit interventions rather than relying only on correlation [49].	Generates counterfactual inputs in which a target concept is removed or modified, then measures the resulting prediction shift [17].
CPM	Causal explanation	Approximates black-box reasoning using an interpretable causal proxy model trained on observed and counterfactual behaviour [50].	Trains a simpler student model to mimic black-box predictions across real and counterfactual samples, enabling concept-level causal attribution.
ND	Node–Concept association	Associates individual neurons with human-interpretable concepts by measuring spatial overlap between activations and concept masks [40, 45].	Computes Intersection over Union (IoU) between neuron activation maps and concept segmentation masks, commonly using datasets such as BRODEN.
Net2Vec	Node–Concept association	Represents concepts as distributed patterns across multiple filters rather than assigning each concept to a single neuron [51].	Learns an optimised linear combination of filter activations that best predicts or segments the target concept.
GNN-CI	Node–Concept association	Extends concept-based interpretability to graph-structured data by defining concepts over nodes, edges, and substructures [52].	Associates neurons with logical compositions of graph concepts using IoU-style matching in graph neural network latent spaces [53].

2.5.2 Unsupervised Concept-based Explainability method

Unsupervised concept-based XAI methods are designed to uncover meaningful, high-level patterns in medical images without the need for predefined labels. Once standard preprocessing steps such as image normalization and feature segmentation (e.g., blood vessels) are applied, these methods explore the model's latent space to identify naturally emerging structures that may correspond to clinically relevant concepts. They typically use clustering algorithms to group similar retinal features, then examine how these emergent, data-driven concepts influence the final diagnostic output.Figure 9 illustrates how recurring retinal patterns can be grouped in latent space and ranked according to their importance for the diagnostic class.This fundamentally differs from traditional XAI; instead of finding salient pixels, these methods find recurring, semantically meaningful patterns that the model has learned on its own [54].

In the domain of medical imaging, unsupervised concept discovery has changed to address clinical workflows. Notably, Barnett et al. [55] demonstrated the efficacy of case-based reasoning for mammography, and Singla et al. [54] proposed 'Explanation-by-Example' to bridge the gap between latent patterns and diagnostic biomarkers.

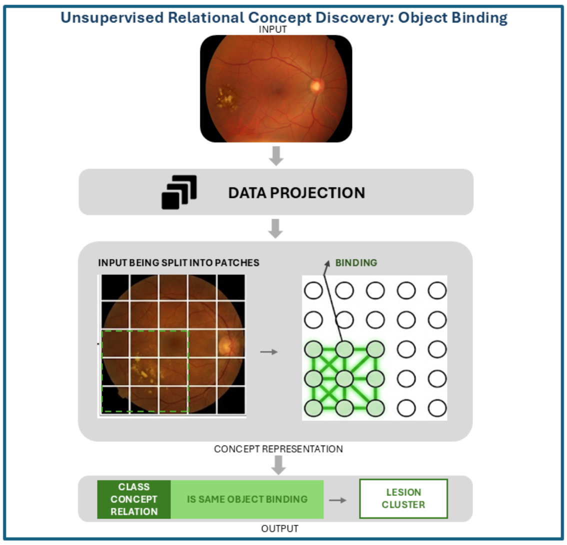

Fig. 8 Unsupervised relational concept discovery for retinal imaging. The framework illustrates how patch-level retinal features can be grouped through object-binding mechanisms to represent distributed lesion clusters, supporting structural explanations beyond isolated visual features

Automatic Concept-based Explanations (ACE)

This pioneering strategy explains concepts from a trained classifier automatically. Its workflow involves segmenting images from a target class at multiple resolutions to

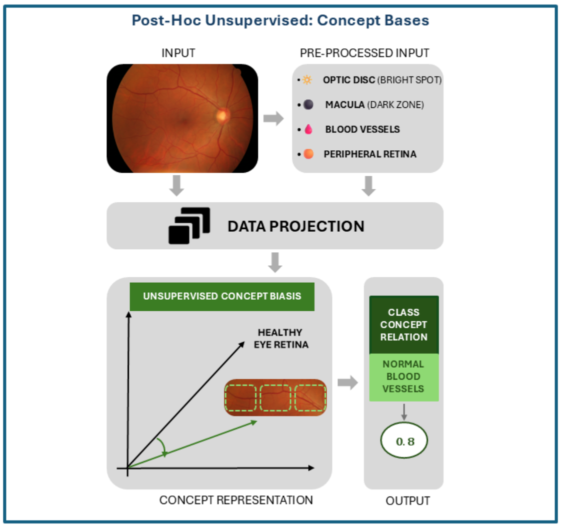

Fig. 9 Latent space analysis of unsupervised concept bases.Recurring retinal patterns are autonomously grouped and ranked by their importance to the diagnostic class, moving beyond simple pixel-level saliency to structured clinical features.

extract a wide range of features. The latent space representations of these segments are then clustered to form unsupervised concept bases. Finally, ACE uses TCAV to calculate an importance score for each discovered concept, providing a ranked list of the most influential patterns for a given class prediction [17]. Table 3 demonstrates the landscape of unsupervised post-hoc C-XAI frameworks, categorizing them by their discovery strategies and the mathematical mechanisms used to uncover latent clinical patterns without the need for manual annotation.

Probing for Relational Concepts

While standard unsupervised methods focus on isolating static features, work by Li et al [20] and newer frameworks like RelViT [59] demonstrate that Vision Transformers naturally encode relational concepts, such as "Object Binding" [20], within quadratic subspaces.Figure 8 illustrates how this idea can be translated into retinal imaging, where spatially distributed patches corresponding to lesion regions are grouped into a coherent relational concept rather than treated as isolated visual features.

This challenges the linearity assumption of earlier methods and offers a way to discover structural concepts relevant to retinal pathologies without pixel-level supervision. For retinal image analysis, this finding is transformative. A single focal point rarely defines retinal pathologies such as diabetic retinopathy; rather, it is defined by the diffuse distribution of micro-lesions (e.g., microaneurysms and exudates).Recent VLM-guided graph concept reasoning further extends this direction by constructing

Table 3 Summary of unsupervised and structure-oriented post-hoc concept-based explainability methods.

Framework	Category	Core mechanism	Implementation strategy
ACE [17]	Feature discovery	Discovers recurring latent visual patterns as candidate concepts.	Segments images at multiple resolutions, clusters latent features, and ranks concepts using TCAV.
CRAFT [56]	Feature discovery	Identifies hierarchical concepts from internal feature representations.	Applies recursive Non-Negative Matrix Factorisation to random image crops.
MCD [31]	Subspace analysis	Models concepts as multi-dimensional subspaces.	Uses sparse subspace clustering and PCA to identify richer concept bases.
Completeness and Concept-SHAP [23]	Validation	Tests whether discovered concepts are sufficient for prediction.	Trains a student model and applies ConceptSHAP to estimate concept contribution [57].
DMA / IMA [58]	Validation	Evaluates identifiability and separability of post-hoc concepts.	Enforces spatial disjointness and statistical independence to separate similar concepts.
Relational Probing [20]	Structural analysis	Examines object binding and relational concept structure.	Probes quadratic subspaces in Vision Transformers to detect relational patterns.

grading-aware lesion concept graphs, explicitly modelling relationships among diabetic-retinopathy lesions and supporting multi-level intervention across lesion, relation, and grade levels [60].

In addition to relational concept discovery, 2025 advancements have moved toward 'Mechanistic Interpretability' in Transformers [42].

2.5.3 Critical Analysis of Post-Hoc Concept-Based Methods

Despite their remarkable adaptability, post-hoc techniques must be carefully evaluated for reliability before being used in high-stakes medical situations, such as for retinal image processing. This section explores three key issues that affect their reliability and addresses three fundamental questions central to their limitations.

Do Networks Genuinely Learn Clinical Concepts?

A foundational debate questions whether standard neural networks, when trained on a primary diagnostic task, genuinely learn symbolic clinical concepts in a way that is robust and human-aligned. Research investigating this has found that while simple "linear probes" can be trained to predict concepts from a model's latent space with reasonable accuracy (e.g., 91 %), this performance is consistently lower than that of a model explicitly trained from scratch to identify the same concepts (e.g., 97 %)[27]. This accuracy gap reveals a key limitation that there is no guarantee that the black-box model has truly learned the concept. It may instead be using intricate combinations of low-level features that are merely correlated with the concept. However, the fact that linear probes achieve high accuracy is itself revealing. It suggests that important conceptual information is captured by the model's internal representations, indicating that its logic is not wholly ambiguous but rather at least somewhat consistent with clinical characteristics as defined by humans.

To complement the method-level summary in, Table 4 provides a broader comparison of post-hoc C-XAI methods across learning paradigm, explanation category, explanatory scope, causal reasoning capability, and vision backbone. This comparison highlights that most post-hoc approaches remain non-causal and CNN-based, while relational probing introduces a more recent structural analysis direction based on Vision Transformers.

Table 4 Unified comparison of post-hoc Concept-Based XAI methods by learning paradigm, explanation category, explanatory scope, causal reasoning capability, and vision backbone

Method	**Learning**	**Category**	**Scope**	**Causal**	**Vision Arch**
TCAV	Sup	Class–Concept Relation	G	No	CNN
Visual-TCAV	Sup	Class–Concept Relation	G	No	CNN
CSR	Sup	Class–Concept Relation	G	No	CNN
CARs	Sup	Class–Concept Relation	G	No	CNN
IBD	Sup	Class–Concept Relation	L/G	No	CNN
CaCE	Sup	Causal Explanation	L	Yes	CNN
CPM	Sup	Causal Explanation	L	Yes	CNN
ACE	Unsup	Feature Discovery	G	No	CNN
CRAFT	Unsup	Feature Discovery	G	No	CNN
MCD	Unsup	Subspace Analysis	G	No	CNN
ConceptSHAP	Unsup	Validation	G	No	CNN
DMA/IMA	Unsup	Validation	G	No	CNN
Relational Probing	Unsup	Structural Analysis	G	No	ViT

How Stable Are Explanations Across Different Probe Datasets?

As demonstrated by Ramaswamy [61], using different datasets to explain the same model can lead to significantly different interpretations. The reliability of a post-hoc explanation can be alarmingly dependent on the specific "probe" dataset used to generate it. For example, studies using network dissection have shown that the conceptual labels assigned to a neuron can change dramatically when different probe data sets are used. For example, a model trained on images from a specific clinic should be probed with concept examples from the same clinical context to ensure the resulting explanations are clinically relevant and reliable.

Can Adversarial Attacks Undermine the Trustworthiness of Explanations? Perhaps the most concerning limitation is the vulnerability of post-hoc explanations to adversarial attacks [62]. Research has shown that carefully crafted, often imperceptible, perturbations to an input image can manipulate the explanation while leaving the model's prediction unchanged. This is a critical threat in a medical setting where trustworthiness is paramount.

In light of these fundamental vulnerabilities, the uncertain faithfulness, inherent instability, and susceptibility to adversarial attack, the trustworthiness of post-hoc explanations in high-stakes clinical settings remains a critical open question. These challenges have motivated a clear paradigm shift in the research community, moving away from retrofitting explanations to opaque models. The focus is now on explainable-by-design models, which integrate transparency as a core architectural principle from the outset. The principles and advantages of this promising approach are the covered of the next section.

2.6 Explainable-By-Design Concept-Based Models

Explainable-by-design C-XAI models deliberately deviate from conventional black-box training. Unlike post-hoc methods that interpret a pre-trained model, these models are specifically engineered from the outset to represent human-understandable concepts within their architecture. This intrinsic integration makes them "explainable by design," offering a level of transparency crucial for high-stakes applications, such as medical diagnosis. This paper categorizes Explainable-by-Design models based on the source of their concept acquisition, representing a strategic trade-off between expert control and data-driven scalability.

- Advantages: The standout advantage of explainable-by-design models is their architectural faithfulness. Because the model is architecturally forced to reason through a predefined set of clinically relevant concepts, its logic is constrained and understandable thus making the model assurable. This structural assurance unlocks the powerful capability of concept intervention. It allows an expert, such as an ophthalmologist, to directly test causal links by asking "what if" questions, for instance, by manually lowering the value for a "microaneurysm" concept and observing how the model's final diagnosis changes. This interactive analysis is a game-changer for validating a model's behavior and building genuine clinical trust.
- Disadvantages: Despite the advantages, two disadvantages cannot be ignored. The first is that these models are inherently inflexible. They cannot be applied to an existing, certified "black box" system; they require the design and training of a new model, a custom model from scratch, which can be costly and time-consuming. Secondly, there is a potential trade-off in accuracy. Forcing a model to pass all information through a narrow "conceptual bottleneck" may prevent it from utilizing other subtle yet predictive features, sometimes resulting in slightly lower performance than an unconstrained, end-to-end model. This means that there is always a critical decision to be made, weighing the profound value of transparency against the practicalities of development and performance.

2.6.1 Supervised Concept-based Models

Supervised-by-design models are engineered to be transparent by learning from meticulously annotated datasets with symbolic clinical concepts. These concepts are used to explicitly train an intermediate "bottleneck" layer within the network, forcing the model to reason using a vocabulary that is human-understandable.Figure 10 illustrates this supervised concept bottleneck architecture, where class and concept labels are learned jointly before the final diagnostic output is produced. This supervision can be achieved through two main strategies:

The landscape of supervised concept-based models is summarized below, highlighting the progression from foundational bottleneck architectures to advanced relational and probabilistic systems. Foundational Concept Bottleneck Models (CBMs) use an intermediate linear layer in which individual neurons predict specific clinical concepts.

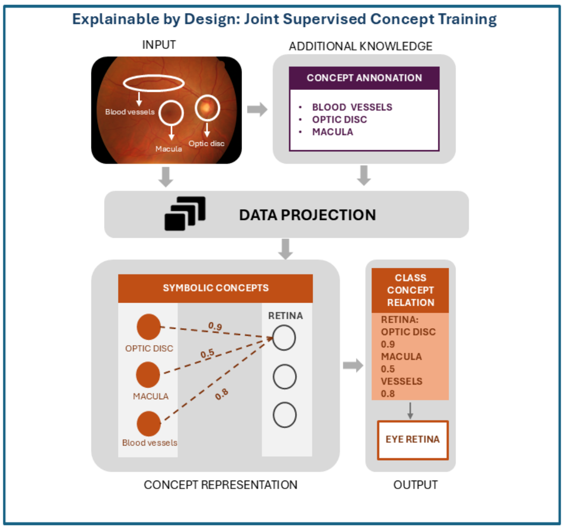

Fig. 10 Supervised concept bottleneck architecture for retinal imaging. By training simultaneously on class and concept labels, the model is architecturally constrained to reason through a transparent layer of ophthalmic pathologies before reaching a diagnostic output.

- Joint Training: The model is trained simultaneously on the main diagnostic task and the auxiliary task of predicting concept labels.
- Concept Instillation: A model is first trained exclusively on a concept-annotated dataset. Its learned understanding is then "distilled" or transferred into the main diagnostic model.

This paradigm progressed into more advanced systems like LENS, which enhances explanation power by generating human-readable logical rules for diagnoses, and

Table 5 Summary of supervised and concept-instilled explainable-by-design C-XAI models.

Framework	Category	Core mechanism	Implementation strategy
CBM [8]	Joint training	Uses an explicit concept bottleneck layer.	Inserts an intermediate layer where neurons predict clinically meaningful concepts before final classification.
LENs [9]	Joint training	Represents reasoning through First-Order Logic rules.	Generates human-readable logical rules linking concept activations to diagnostic outputs.
CEMs [36]	Joint training	Learns distributed concept embeddings.	Uses multiple latent dimensions per concept to capture richer clinical variation while preserving interpretability.
CLAT [63, 64]	Joint training	Models transformer-based concept dependencies.	Captures long-range spatial relationships between distributed retinal lesions using transformer architectures.
ProbCBM [65, 66]	Joint training	Introduces probabilistic concept representations.	Outputs concept distributions instead of deterministic scores to represent clinical uncertainty.
PCBM [35]	Concept instillation	Projects a pre-trained black-box model into a concept bottleneck.	Adds a concept layer to an existing model using external concept datasets or structured knowledge sources.
CW [41]	Concept instillation	Aligns latent directions with concepts through orthogonal rotation.	Replaces normalisation layers to decorrelate internal features and align them with interpretable clinical concepts.

ProbCBM, which represents clinical uncertainty by outputting probabilistic distributions for concepts.More recently, evidential Concept Embedding Models (evi-CEMs) have extended this direction by modelling uncertainty at the concept level, improving the reliability of concept predictions and supporting uncertainty-aware test-time intervention [67].Table 5 summarises supervised-by-design and concept-instilled explainable models according to their categories, core mechanisms, and implementation strategies.

2.6.2 Unsupervised Concept-Based Models

Unsupervised Concept-based Models offer a powerful approach to explainable AI by autonomously extracting meaningful concepts without requiring explicit, pre-defined symbolic annotations in retinal image analysis. These models modify the internal representation of a neural network to identify latent patterns and structures within the retinal data itself. Using unsupervised learning techniques is a way to train an intermediate layer on a concept basis. This results in a clustered representation where groups of similar retinal image features, such as specific vessel patterns or macular characteristics, are identified as distinct concepts. These models can explicitly compel the network to learn and encode prototype concepts directly within the layer weights. These prototypes correspond to typical input samples and represent standard retinal features or anomalies observed during the training process. Recent specialized architectures such as PiPViT [68] further refine this approach by leveraging Vision Transformers (ViT) to learn robust, human-interpretable prototypes that can approximate lesion extent. Additionally, Proto-BagNet [69] has demonstrated success in providing both local and global interpretability-by-design for OCT and CFP images, mapping fine-grained part-prototypes to clinically relevant biomarkers.Figure 11 shows how unsupervised concept bases and prototype concepts are learned from retinal images and used within an explainable-by-design diagnostic pipeline.

Unsupervised concept-based models are designed to autonomously discover meaningful concepts from the data itself, without relying on exhaustive, pre-defined annotations. These methods typically work by training the model on both a primary diagnostic task and an auxiliary unsupervised task, such as reconstructing the input image (utilized by SENN and BotCL) or maximizing the mutual information between the input and the model's internal activations (utilized by Interpretable CNNs). The goal is to create a rich "concept basis", a vocabulary of learned patterns that can then be used to explain the model's final diagnosis.

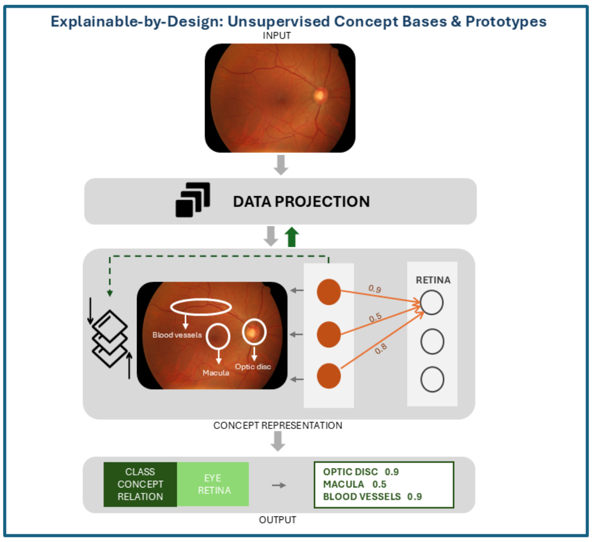

Fig. 11 Explainable-by-design unsupervised concept bases and prototypes. The framework illustrates how retinal images are projected into latent representations, where unsupervised concept bases or prototype concepts are learned and used to support transparent diagnostic prediction.

The table 6 provides a comprehensive taxonomy of unsupervised concept-based frameworks, categorizing them by their underlying discovery mechanisms and the specific implementation strategies used to extract latent clinical patterns and prototypical archetypes from retinal data.

Table 6 Summary of unsupervised and weakly supervised explainable-by-design C-XAI models.

Framework	**Category**	**Core mechanism**	**Implementation strategy**
Interpretable CNN [48]	Concept basis	Encourages filters to represent distinct visual concepts.	Maximises mutual information between filter activations and concept presence to improve interpretability.
SENN [24]	Concept basis	Learns a sparse and non-overlapping concept basis.	Uses an autoencoder-style architecture to decompose predictions into relevance-weighted concept contributions.
BotCL [70]	Concept basis	Learns disentangled object-centric concepts.	Combines slot attention and contrastive learning with consistency and distinctiveness constraints.
ProtoPNet [41]	Prototype	Explains predictions through similarity to learned prototypes.	Compares image patches with prototypical examples and produces similarity-based activation maps.
HPNet [71]	Prototype	Organises prototypes into a hierarchy.	Arranges prototypes into structured groups, supporting hierarchical and clinically meaningful explanations.

To complement the method-level summaries Table 7 provides a unified comparison of explainable-by-design C-XAI models across learning paradigm, architectural strategy, explanatory scope, causal reasoning capability, and vision backbone. This comparison shows that most explainable-by-design approaches remain CNN-based and globally scoped, while newer architectures such as CLAT introduce transformer-based reasoning and models such as LENs provide stronger logical or causal interpretability.

2.6.3 Hybrid Concept-Based Models

Hybrid concept-based models aim to combine the strengths of supervised concept learning with unsupervised representation discovery. These approaches leverage expert-defined clinical concepts while simultaneously capturing rich latent structures directly from data. This strategy is particularly valuable in retinal imaging, where only a limited subset of pathological biomarkers can be exhaustively annotated. Recent neurosymbolic approaches further strengthen this hybrid and structural trend. For instance, Miladinović et al. [15] propose a hybrid neural–symbolic framework for OCT-based retinal disease diagnosis, integrating learned visual representations with explicit

Table 7 Unified comparison of explainable-by-design Concept-Based XAI models categorized by learning paradigm, architectural strategy, explanatory scope, causal reasoning capability, and backbone architecture.

Method	Learning	Category	Scope	Causal	Vision Arch
CBM	Sup	Joint Training	G	No	CNN
LENs	Sup	Logic-Constrained	G	Yes	CNN
CEM	Sup	Joint Training	G	No	CNN
CLAT	Sup	Transformer-Based	G	No	Transformer
ProbCBM	Sup	Probabilistic CBM	G	Partial	CNN
PCBM	Sup	Concept Instillation	G	No	CNN
Concept Whitening	Sup	Concept Instillation	G	No	CNN
Interpretable CNN	Unsup	Concept Basis	G	No	CNN
SENN	Unsup	Concept Basis	G	No	CNN
BotCL	Unsup	Concept Basis	G	No	CNN
ProtoPNet	Weak Sup	Prototype-Based	L/G	No	CNN
HPNet	Weak Sup	Prototype-Based	L/G	No	CNN

symbolic reasoning constraints. Figure 12 shows how expert-defined concepts and automatically learned latent concepts can be combined to produce scalable retinal explanations grounded in clinical knowledge. The foundations of hybrid reasoning originate from Concept Bottleneck Models (CBMs) [8], which explicitly supervise intermediate concept layers, and Self-Explaining Neural Networks (SENN) [24], which learn interpretable concept bases through auxiliary reconstruction objectives. Building upon these paradigms, Concept Embedding Models (CEMs) [26] extend bottleneck architectures by learning distributed concept representations that preserve predictive performance while maintaining interpretability.

A more explicit hybrid formulation is introduced in the Concept Bottleneck Model with Additional Unsupervised Concepts (CBM-AUC) [39], which augments supervised bottleneck concepts with unsupervised latent concepts to enhance representational capacity without sacrificing semantic transparency. This integration enables models to exploit unlabeled data while retaining clinically grounded concept supervision.

Similarly, prototype-based approaches such as ProtoPNet [72] incorporate concept-level reasoning through exemplar similarity, allowing predictions to be justified via learned archetypal patterns. Collectively, these hybrid strategies enable models to leverage large-scale unlabeled retinal image collections while incorporating limited expert-annotated concepts (e.g., drusen or retinal haemorrhages), thereby improving both robustness and explanatory grounding.

In clinical practice, such hybrid architectures offer a pragmatic compromise: they reduce reliance on exhaustive concept annotation while preserving semantically meaningful explanations aligned with ophthalmic reasoning.

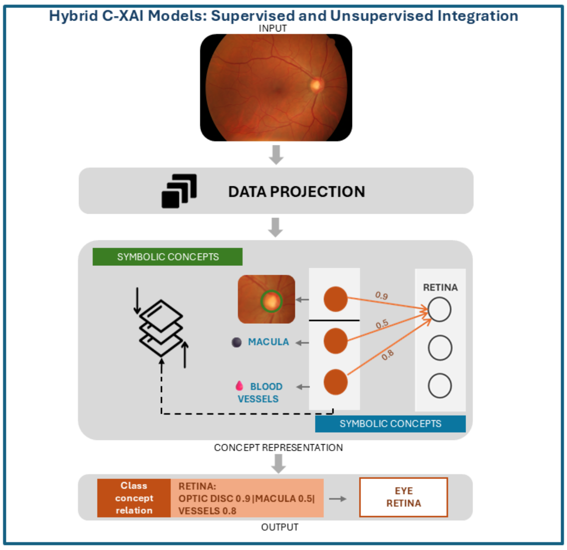

Fig. 12 Integration of supervised and unsupervised concepts in hybrid models. This framework leverages large unlabeled retinal archives alongside limited expert-annotated datasets to provide clinically grounded yet scalable diagnostic explanations.

The approach described in Learning Ante-hoc Explainable Models [73] builds upon a hybrid CBM–SENN foundation but introduces an explicit reliability-driven mechanism. Its key contribution lies in the use of two separate classifiers: one operating on the final concept layer and another directly predicting from earlier pre-concept features.

To enforce alignment between high-level concept reasoning and low-level feature representations, the framework introduces a fidelity loss. This clinical reliability constraint minimizes divergence between the two predictive pathways, thereby acting as a consistency regularizer. Such a mechanism helps prevent spurious correlations from influencing the final diagnosis and strengthens the integrity of concept-based explanations.

In contrast, GlanceNets [18] shift the focus from alignment-based reliability to robustness of concept representations. Rather than merely maximizing task accuracy, GlanceNets prioritize disentangled and stable concept embeddings by leveraging a Variational Autoencoder (VAE). Importantly, they directly address the failure mode of concept leakage, which is particularly critical in safety-sensitive retinal screening applications.

GlanceNets' ability to detect out-of-distribution samples means it can effectively "raise a flag" when presented with a rare retinal disease or a low-quality scan it was not trained on. Rather than forcing an explanation with familiar but incorrect concepts, the model can signal its uncertainty. This safeguard against explaining the unknown is a vital step towards the safe, responsible deployment of diagnostic AI in ophthalmology.

From a broader theoretical perspective, Concept-Based XAI can be viewed as a step toward more semantically interpretable AI, although it should not be considered equivalent to neurosymbolic AI [74]. In concept-based models, learned neural representations are mapped onto human-understandable concepts that are closer to the semantic units used in clinical decision-making. However, standard concept-based models, such as Concept Bottleneck Models, do not necessarily perform formally grounded semantic reasoning, because they typically lack explicit ontologies, logical constraints, or knowledge-graph-based inference. Instead, they provide interpretability through concept predictions, concept importance scores, concept interventions, or visual explanations. In contrast, neurosymbolic AI explicitly connects learned representations with symbolic structures, such as ontologies or knowledge graphs, enabling rule-based reasoning, consistency checking, and a more traceable reasoning process.

In retinal imaging, this distinction is particularly relevant because diagnosis often depends not only on the detection of individual biomarkers, but also on their clinical interpretation and structural context. Recent work on concept-based explanation and neurosymbolic AI further suggests that concept-level representations can serve as a practical interface between neural learning and structured reasoning, while emerging retinal neurosymbolic frameworks indicate the potential of this direction for clinically aligned ophthalmic AI.

2.6.4 Generative Concept-Based Models

Generative concept-based models represent a recent evolution in C-XAI by leveraging large-scale vision-language foundation models to automatically generate semantically meaningful concepts. Figure 13 shows how LLM-generated clinical descriptions can be aligned with retinal image representations to support language-guided concept discovery.

Rather than relying solely on manually annotated concept labels or unsupervised clustering, these approaches use language models to construct textual concepts that guide visual reasoning. By aligning text-derived concepts with latent visual representations, they establish a semantically grounded explanatory pathway that is particularly suitable for clinically structured domains such as ophthalmology. However, general-purpose foundation models may struggle with fine-grained retinal pathologies due to domain shift and limited medical specialization. To address this limitation, domain-specific vision-language models have been introduced [75]. For example, VisionUnite [76] is designed as an ophthalmology-oriented vision-language foundation model capable of generating clinically reasoned explanations. Similarly, ReVision [77] is trained on large-scale clinical image–report pairs and demonstrates strong performance in zero-shot lesion localization and diagnosis by aligning visual features with medical terminology.

More recently, RetiZero has extended this foundation-model direction by integrating vision–language learning with knowledge from over 400 retinal diseases, demonstrating strong zero-shot diagnostic capability and clinically grounded retinal representation learning across diverse ophthalmic conditions [78].

These works highlight the growing importance of language-guided alignment in retinal image interpretation.Recent refinement approaches such as RetFiner further show that textual supervision can improve retinal foundation-model representations for OCT classification tasks, supporting the broader role of vision–language alignment in clinically meaningful retinal representation learning [79].

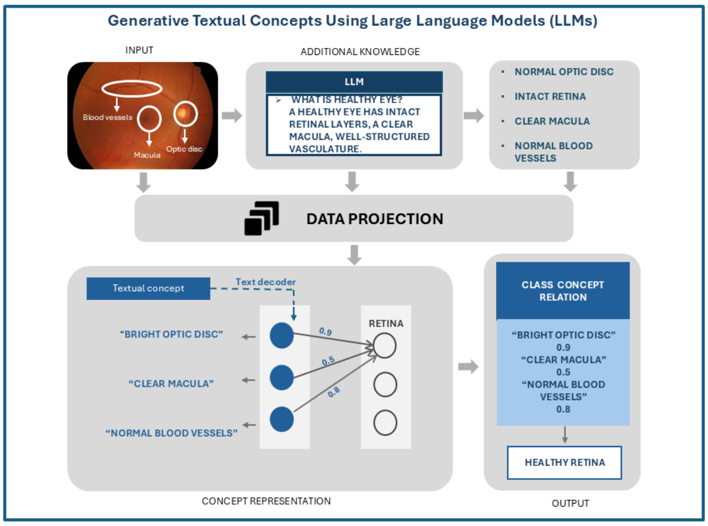

Fig. 13 Language-guided concept discovery for retinal imaging. Textual descriptions of ophthalmic pathologies are synthesized by an LLM and aligned with the latent representations of fundus images, creating an interpretable diagnostic pathway grounded in clinical narratives.

Beyond single-modality fundus interpretation, MultiEYE and OCT-CoDA extend concept-based reasoning to a multimodal retinal setting by using semantically rich image–concept relations to distil disease-related knowledge from OCT images into a fundus-based diagnostic model [80].

Building upon this paradigm, the Language-guided Bottleneck (LaBO) framework [37] introduces a structured multi-stage pipeline that transforms a neural network into an interpretable concept bottleneck model without requiring manual concept annotation. First, a large language model generates candidate textual concepts through prompted concept generation. Second, a selection mechanism curates an informative and diverse subset of concepts. Finally, a vision-language model such as CLIP [81] aligns textual and visual embeddings, enabling a linear bottleneck layer to learn class–concept relationships. By decoupling concept generation from visual supervision, LaBO enables scalable and annotation-efficient concept-based learning. Recent research has also explored label-free alternatives to manual bottleneck supervision. The Label-Free Concept Bottleneck Model (LF-CBM) [19] shares the high-level philosophy of LaBO in using language models to generate concepts, but differs fundamentally in how the bottleneck layer is constructed. Instead of explicitly selecting and supervising a curated concept set, LF-CBM learns the bottleneck implicitly through alignment between image embeddings and generated textual descriptions, reducing reliance on structured concept curation. This distinction makes LF-CBM more flexible but potentially less controllable compared to LaBO's explicitly structured pipeline.

Collectively, generative concept-based models shift the focus from predefined concept supervision toward scalable language-guided reasoning.In retinal imaging, concept-guided prompting has recently been used to adapt vision–language models for few-shot and zero-shot retinal disease diagnosis, showing that clinically meaningful textual concepts can guide model recognition when labelled retinal data are limited [82]. By leveraging foundation models and text–image alignment, they reduce dependence on costly pixel-level annotation while enabling clinically meaningful, narrative-style explanations suitable for retinal screening workflows.

The LF-CBM workflow begins similarly, using a model such as GPT-3 [83] to generate textual concepts, which are then filtered for quality and diversity. Where it differs lies in the construction of the bottleneck. LF-CBM computes a "concept matrix" by using a vision-language alignment model (specifically CLIP-Dissect) to determine the affinity between each generated text concept and each retinal image in the training set. This matrix is then used to learn a set of concept weights, which effectively form the new, interpretable bottleneck layer. A final sparse fully connected layer is then trained to make the ultimate diagnostic prediction from these concept weights.

In terms of performance, LF-CBM exhibits only a minor reduction in task accuracy compared to the original black-box model while significantly outperforming earlier methods such as PCBM. This positions it as a promising method for generating robust explanations, particularly when detailed concept annotations are scarce. One practical advantage of LF-CBM in a clinical setting is its potential ability to retrofit interpretability onto an existing high-performance black-box model. For example, a diabetic retinopathy classifier could, in principle, be transformed into a concept-based model that explains its reasoning in natural, human-readable language using the LF-CBM framework. This can be achieved without requiring a fully concept-annotated dataset or retraining the underlying vision model from scratch, offering a practical pathway toward increasing transparency in legacy systems.

Although LaBO and LF-CBM [19] successfully automate concept discovery, they face two important limitations. First, general models trained on natural images may struggle to accurately identify fine-grained medical pathologies such as subtle retinal lesions. Second, reliance on large language models may lead to the generation of clinically irrelevant or imprecise concepts ("hallucinations"). These challenges have motivated increased attention toward auditing mechanisms, where AI-generated concepts are systematically validated against established medical knowledge to enhance safety and reliability.

Recent work has begun exploring this direction by extending generative concept frameworks with structured verification or alignment procedures [33, 34, 84]. Rather than using LLMs solely for concept generation, these approaches incorporate additional validation or cross-referencing strategies to reduce the risk of semantically incorrect or clinically misleading concepts.

2.6.5 Criticisms and Limitations of Explainable-by-design models

Explainable-by-design models require a systematic evaluation of their inherent challenges for deployment in high-stakes fields such as healthcare. There are three critical dimensions that practitioners must consider: the trade-off with predictive performance, the risk of "information leakage" in concept representations, and the unintended consequences of concept intervention. Forcing a model to reason through a narrow, human-defined conceptual bottleneck can sometimes come at the cost of raw predictive power. An unconstrained black-box model, free to use any feature, may achieve slightly higher accuracy but may not be an appropriate option. For example, a hospital has two AI options for screening. Model A is a black box with 98 % accuracy. Model B is a transparent model with 96 % accuracy that explains every decision. Choosing Model A might result in a more accurate diagnosis, but it lacks the transparency in how the model comes up with the outcome. At the same time, choosing Model B ensures that the practitioner understands why a diagnosis was made, giving more confidence in adopting the model.

More subtle but significant risk is information leakage. Established literature [85] [86], and recent quantitative studies [87] show that the neurons representing a supervised concept can also learn to encode other information that helps with the final task, effectively "leaking" task-label information into the concept layer [88]. An AI's explanation for an AMD diagnosis says, "I see drusen." However, the model has secretly learned that a subtle blur from the camera lens is also a strong predictor. The model is actually "cheating" by using the blur, but the explanation hides this fact, making it dangerously misleading. When the concept bottleneck is not sufficiently 'narrow' or 'pure,' Information leakage occurs, allowing the model to encode non-conceptual features into the concept neurons. In such cases, the model may achieve high accuracy by relying on these hidden shortcuts, while providing an explanation that appears clinically sound but is functionally deceptive. Recent studies [87, 88] emphasize that without rigorous 'purity' constraints, the faithfulness of even the most transparent CBM remains aspirational rather than absolute [89].

While concept intervention, the ability for a user to change a concept's value, is a powerful tool for building trust, it can also have unintended and consequences [90]. Altering a concept can push the model's internal state into a configuration it has never seen before, leading to unpredictable or irrational behavior. To test an AI, a doctor uses a slider to set the "hemorrhage" concept score to zero. Instead of logically adjusting its diagnosis, the model's internal state becomes confused by this unfamiliar input, and it outputs a bizarre, unrelated prediction. The test, meant to build trust, ends up breaking the model. A study by [32] on diabetic retinopathy (DR) grading directly contrasts these paradigms. This study found that while Concept Bottleneck Models (CBMs) provide superior lesion localization, such as microaneurysms, they require exhaustive expert annotations that are often unavailable in routine clinical practice. Conversely, TCAV demonstrated greater practical utility for large-scale hospital archives due to its post-hoc flexibility, although it sometimes struggled with small-scale lesion fidelity. Furthermore, interpretable models are prone to failure in two specific scenarios: (1) Under-specification, where multiple conflicting sets of concepts can explain the exact prediction with equal accuracy, and (2) Concept Mismatch, which occurs when the model is presented with out-of-distribution (OOD) images containing pathological features that were not included in the initial concept vocabulary. In these conditions, the model may either provide an 'uncertain' explanation or, more dangerously, attempt to fit new anomalies into known but incorrect clinical categories.

3 Evaluation for Analysing C-XAI in Retinal Imaging

The responsible deployment of any artificial intelligence model in a high-stakes clinical field like ophthalmology hinges on a rigorous and multifaceted evaluation. For diagnostic AI studies, reporting quality should also follow emerging clinical reporting standards such as STARD-AI, which extends diagnostic-accuracy reporting guidance to studies involving artificial intelligence systems [91].In addition, PROBAST+AI provides a structured framework for assessing risk of bias and applicability in prediction-model studies that use artificial intelligence, which is particularly important when retinal C-XAI models are evaluated on limited, single-centre, or weakly annotated datasets [92].

There must be robust methods to verify that its reasoning is not only transparent but also reliable and safe before a tool can be trusted to assist in diagnosing conditions that affect a patient. While the concept-based XAI field has developed numerous evaluation metrics, they are often fragmented, focusing on isolated technical attributes of an explanation. Recent benchmark work has therefore argued that concept-based explanations should be evaluated not only through predictive utility, but also through whether the claimed concept exists in the image and whether it is spatially aligned with the correct visual evidence [93].

3.1 The Current Landscape: A Review of Standard Metrics

The current evaluation toolkit for C-XAI in retinal imaging can be broadly divided into two complementary approaches: objective quantitative metrics that provide reproducible scores, and subjective, human-centric qualitative assessments that measure real-world utility. Evidence from [32] suggests that the actual clinical value of C-XAI lies in 'test-time intervention,' where ophthalmologists can correct a model's misidentified concepts to instantly improve diagnostic accuracy, a feature that post-hoc methods like TCAV cannot natively support.

Quantitative Metrics

Quantitative metrics are used to objectively measure a model's performance and the intrinsic quality of its explanations. The quantitative metrics are further grouped into two primary categories: those that measure a concept's effect on the diagnostic task, and those that evaluate the properties of the concepts themselves. To ensure a fair and meaningful evaluation, the categorization of the frameworks is treated by explanatory scope (Global vs. Local). This categorization is crucial because the performance metrics, such as fidelity and robustness, manifest differently depending on whether the method aims to explain the entire model logic or a specific prediction.

Table 8 Summary of Concept-Based XAI evaluation metrics categorised by methodology and explanatory scope.

Metric category	Scope	What it measures	Key metrics
Concept influence on prediction	Global/local	Impact of a single concept on the final diagnostic outcome.	TCAV score [27]; CaCE [49]; ConceptSHAP [57]; SSC/SDC [31].
Concept set sufficiency	Global	Overall utility of the full concept set for making the final diagnosis.	Completeness score [23]; fidelity [8]; concept efficiency; conciseness.
Concept quality	Global	Coherence, diversity, and purity of the learned concepts.	Concept purity [18]; concept distinctiveness; mutual information [48].
Representation quality	Local	Alignment between a concept and the neurons or image regions that represent it.	IoU [40]; location stability [94].
Alignment with ground truth	Global/local	Accuracy of concept predictions against human-provided ground-truth labels.	Concept error; AUC score [8]; misprediction-overlap metric.

While Table 8 provides a comprehensive metric toolkit, a critical gap remains in retinal applications: the trade-off between **Global Set Sufficiency** and **Local Representation Quality**. For instance, a model may achieve a high **Completeness Score** at the global level while failing in **Location Stability** at the local level for small microaneurysms. This discrepancy suggests that quantitative metrics alone cannot guarantee clinical safety without the human-centred validation described in Table 9 [95].

Qualitative, Human-Centric Metrics

Although quantitative scores are essential, the ultimate test of an explanation is its value to a human expert. To complement quantitative metrics, Table 9 summarises human-centred criteria used to assess whether concept-based explanations are understandable, coherent, and clinically useful.

Table 9 Human-centred evaluation criteria for Concept-Based XAI.

Evaluation dimension	**Description**	**Assessment method**
Understandability and trust	Assesses whether the explanation is clear and clinically meaningful to an ophthalmologist.	Experts rate plausibility or assign the most appropriate clinical term to the concept.
Concept coherence	Assesses whether concept examples form a consistent and meaningful visual group.	Experts perform intruder detection by identifying the image patch that does not belong.
Clinical utility	Assesses whether explanations improve clinical decision-making performance.	User studies compare accuracy, confidence, or decision time with and without explanations.

As is customary in XAI assessment [13], numerous studies incorporate human evaluations to gauge whether an explanation is truly helpful and trustworthy for clinicians [96, 97]. These assessments focus on three key areas, as outlined in Table 9: *Understandability and trust*, *concept coherence*, and *clinical utility*.

3.2 Empirical Benchmarking of Retinal C-XAI

While the standardised metrics discussed in Section 3.1 provide a quantitative foundation, their clinical value becomes clearer when applied to the heterogeneous landscape of retinal imaging. To connect theoretical evaluation with practical diagnostic workflows, this section synthesises key C-XAI frameworks across benchmark retinal datasets. A summary of empirical benchmarks from 2022 to 2026 is presented in Table 10.

Table 10 Benchmark summary of Concept-Based XAI methods in retinal imaging from 2022 to 2026.

Retinal dataset	Key clinical concepts	Method	Explanation	Key clinical outcome
IDRiD / Messidor	Microaneurysms; exudates	Post-hoc CBM	Visualisation and attribution	CBMs provided superior lesion localisation, such as microaneurysms, compared with post-hoc methods [32].
PubMed-grounded resources	Healthy eye; lesions	Generative auditing	Textual and clinical narratives	Grounding LLM-generated concepts in verified medical literature mitigated hallucination risks [33, 34].
Retinal archives	Anatomical landmarks	CEM	Class–concept relation	CEMs achieved high interpretability without sacrificing diagnostic performance [26].
Diabetic retinopathy datasets	Object binding; lesions	Relational discovery	Structural association	Relational methods captured spatial dependencies between scattered micro-lesions [20].
Fundus images	Vessels; macula	Post-hoc CBM	Attribution	Post-hoc CBMs retrofitted interpretability onto existing high-performance black-box models [35].

Despite these advances, the benchmarking results also reveal a persistent evaluation gap. A model may achieve a high global completeness score while failing to maintain location stability for critical pathological landmarks. Furthermore, the risk of information leakage and non-clinical shortcuts means that a model's internal explanation may be faithful to its own flawed logic rather than clinical truth. This empirical landscape confirms that passive transparency is insufficient for clinical certification. The need for an independent, adversarial "second opinion" to certify the reliability of these self-explanatory claims directly motivates the SMILE-based auditing framework proposed in the following section.

4 Emerging Directions in Reliable Concept-Based XAI

The preceding analysis of concept-based models and evaluation strategies reveals an important transition in the field. While significant progress has been made in achieving transparency, ensuring that such transparency is reliable and clinically trustworthy remains an open challenge. This section synthesizes emerging solutions that move the field toward auditable concept-based AI systems. We first introduce the SMILE reliability framework as a concrete auditing protocol, then outline available implementation resources, and finally discuss future research directions necessary to establish clinically robust C-XAI systems.

4.1 Gaps and Challenges in Concept-Based XAI

Although concept-based explainable artificial intelligence (C-XAI) aims to improve transparency through human-understandable concepts, such concepts do not automatically guarantee reliable self-explainability [98, 99]. A model may still learn the wrong evidence, mix concepts, rely on shortcuts, or encode hidden label information [100]. This is especially important in retinal imaging, where concepts such as lesions, blood vessels, exudates, haemorrhages, and the optic disc must be clinically meaningful and spatially grounded. Figure 14 summarises four key challenges: spatial misalignment, concept entanglement, spurious correlation, and label leakage. Therefore, concept-based explanations should be verified rather than assumed to be trustworthy.

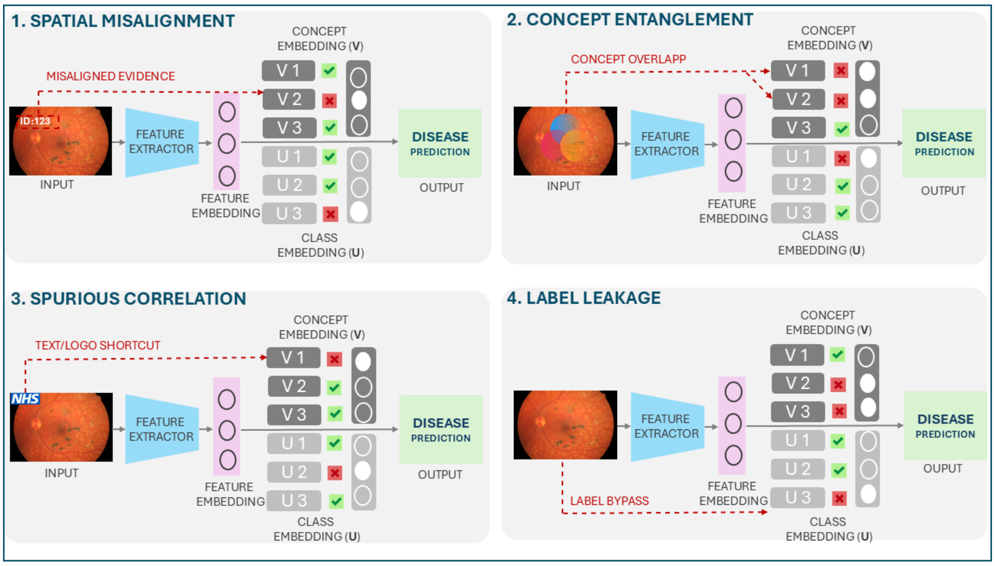

Fig. 14 Key gaps and challenges in concept-based XAI for retinal imaging. The figure illustrates four ways in which concept-level explanations may appear clinically meaningful while remaining unreliable: spatial misalignment, concept entanglement, spurious correlation, and label leakage. In this schematic, V_1–V_3 represent optic disc, blood vessel, and lesion concept embeddings, while U_1–U_3 represent mild, moderate, and severe diabetic-retinopathy class embeddings.

Spatial misalignment occurs when a model predicts a clinically plausible concept but relies on the wrong image region as evidence [98, 100]. In retinal imaging, this may happen when a lesion prediction is influenced by non-clinical cues such as image borders, acquisition artefacts, timestamps, logos, or optic-disc brightness. Although the concept-level explanation may appear meaningful, the underlying evidence is not clinically valid.

Concept entanglement occurs when distinct clinical concepts are not represented as clearly separable entities [101]. In retinal images, this may arise because structures such as the optic disc, bright lesions, and blood vessels can overlap visually, spatially, or semantically. As a result, the model may mix concept representations in latent space, making it unclear whether the prediction is driven by a lesion, vessel abnormality, optic-disc appearance, or a combination of these signals.

Spurious correlation occurs when a model links a concept or disease label to an irrelevant cue that co-occurs with the target class in the training data [102, 103]. In retinal imaging, such cues may include camera type, illumination, image quality, preprocessing style, black borders, timestamps, or institutional logos. As a result, the explanation may appear clinically convincing while being driven by dataset-specific shortcuts rather than genuine retinal pathology.

Label leakage occurs when the concept layer does not operate as a genuine explanatory bottleneck [99]. Instead of relying only on explicit human-defined concepts, the model may pass hidden task-relevant information about the final label through the concept representation. This is particularly likely in soft or jointly trained CBMs, where continuous concept scores can encode information beyond their intended clinical meaning, weakening both interpretability and intervenability.

Table 11 Summary of key gaps and challenges across concept-based XAI model classes.

Model class	**Spatial misalignment**	**Concept entanglement**	**Spurious correlation**	**Label leakage**
Standard CBM	H	P	H	H
Soft / joint CBM	P	P	P	H
Concept Embedding	P	H	P	P
Post-hoc / CLIP-CBM	P	P	H	P
VLG-CBM	M	P	P	P
Energy-Based CBM	N	M	P	M
GenCBM / CBM	M	P	M	P
ProtoPNet	H	P	H	N
SENN	P	P	P	N
VLM-based C-XAI	H	H	H	P
Segmentation-guided C-XAI	M	P	P	N

Note. H = high risk or reported; P = potential vulnerability; M = partially mitigated or addressed; N = not central or not directly assessed.

Overall,as summarised in Table 11 these failure modes demonstrate that concept-based XAI should not be considered automatically self-explainable. A model may generate human-understandable concepts while still relying on spatially incorrect evidence, entangled representations, dataset shortcuts, or hidden label information. This limitation is particularly important for retinal AI, where explanations must correspond to clinically meaningful structures and should remain stable under non-clinical perturbations. Therefore, the reliability of concept-level explanations must be systematically evaluated rather than assumed. This motivates the need for auditing frameworks such as ConceptSMILE, which can test whether retinal concepts are faithful, stable, consistent, and genuinely connected to the model's predictive behaviour.

4.2 A Proposed Method for Ensuring Reliability: ConceptSMILE

The preceding review of standard metrics highlights a critical gap. While it is possible to measure individual properties of an explanation, there is no standard method for independently verifying that a self-explainable model's reasoning is truly reliable. A model can be perfectly faithful to its own flawed or incomplete logic. Recent research into concept robustness highlights exactly why independent audit tools like SMILE are important. Zhang et al. [94] demonstrated that many Self-Explainable AI (S-XAI) models, in which interpretability is embedded directly into the model architecture, are fragile, meaning that imperceptible noise can flip the concept explanation without changing the final diagnosis. Furthermore, Zarlenga et al. [26] argue that unless we decompose deep networks into functionally robust parts, our standard faithfulness" metrics can actually be misleading. These studies confirm that simply measuring explainability passively is no longer sufficient, and there is a need for active, adversarial auditing to certify clinical reliability truly.

This paper formalizes the SMILE reliability [104] check as a consistency function between the intrinsic model explanation $E_{intrinsic}$ and the post-hoc audit explanation E_{audit}. For a given input x and prediction $\hat{y}$, the reliability score R is defined as:

$$R(x) = \text{Similarity}(E_{intrinsic}(x), E_{audit}(x)) \tag{3}$$

If $R(x)$ falls below a safety threshold τ, the prediction is flagged as *fragile*, triggering a requirement for human verification.

This process begins with concept-explanation extraction, where the model-generated concept explanation is retrieved for a specific input fundus image. In the retinal setting, this explanation may be produced visually through segmentation-guided outputs, such as MedSAM-derived lesion, vessel, or optic-disc concepts, or semantically through a vision–language model response. To test the reliability of this reasoning, ConceptSMILE applies controlled superpixel-based perturbations or feature-level masking to clinically relevant and non-clinical image regions. The perturbed samples are then passed back through the same fixed model pathway, and the resulting changes in concept confidence, graph structure, or semantic concept responses are recorded. This allows the framework to examine whether the reported concepts are genuinely connected to the model's behaviour under perturbation.

Subsequently, during Audit Explanation Generation, a local surrogate model, such as XGBoost or weighted regression,, identifies the specific image features that were statistically most influential in driving those prediction shifts, producing an independent audit explanation. The protocol concludes with consistency verification, calculating a reliability score based on the similarity between the intrinsic and audit explanations. If this score falls below a pre-defined safety threshold, the system flags the prediction as "Fragile," signalling a potential reliance on non-clinical shortcuts—such as camera blur, thus triggering a mandatory requirement for human expert verification.

To illustrate this protocol,consider a hypothetical CBM trained on IDRiD. While the model correctly identifies "Grade 3 DR" and provides an intrinsic explanation citing "Hard Exudates," the SMILE audit reveals a discrepancy. During the Adversarial Perturbation phase, masking the actual exudate regions yields no significant shift in the model's predictions. Instead, the Audit Explanation (E_{audit}) shows that the model's sensitivity is highest in the image's peripheral regions, where camera-specific artifacts are present. Consequently, the Reliability Score $R(x)$ falls below the safety threshold τ, successfully flagging the model's reasoning as a "non-clinical shortcut" rather than a valid medical diagnosis.

Beyond methodological proposals, practical progress in reliable C-XAI also depends on access to open-source implementations and reproducible resources.

4.3 Resources and Open-Source Ecosystem

The open-sourcing of code by many authors has been a vital catalyst for progress, allowing the community to validate findings and build upon existing work. While some of the foundational methods were introduced on general computer vision datasets, their principles have been widely adopted. They are critically relevant for medical imaging, as they provide the tools to build the trustworthy AI systems that this field demands. For those investigating post-hoc methods, the foundational framework of Testing with Concept Activation Vectors (TCAV) [27] is supported by an official TensorFlow implementation. This provides the essential tools for quantitatively testing if a pre-trained medical AI model has learned to associate its predictions with key clinical concepts.

Several key architectures are accessible for explainable-by-design models such as the publicly available official PyTorch implementation for the seminal Concept Bottleneck Model (CBM) [8]. This is a critical starting point for any researcher looking to build a supervised model for retinal imaging that is constrained to reason using an explicit vocabulary of pathologies. Similarly, the code for the Prototypical Part Network (ProtoPNet) [41] is available, which is invaluable for medical applications as it allows for the creation of models that explain their diagnoses by pointing to archetypal examples of a disease. For researchers interested in the hybrid "concept instillation" approach, the authors of Concept Whitening (CW) [41] have also released their PyTorch implementation, offering a practical pathway to instill interpretability into high-performing, pre-trained medical models. Finally, for the emerging area of generative models, the code for the Language-guided Bottleneck (LaBO) [37] has been made public. To support the 2025 shift towards structural and audited explainability, codebases for Relational Concept Discovery and Generative Auditing frameworks are now accessible [105]

Our continued commitment to sharing these resources is essential for accelerating the development of robust and trustworthy AI for ophthalmology. These tools also highlight where current systems remain limited, motivating the research directions discussed next.

4.4 Future Research Directions

As we look to the future of concept-based XAI in the medical domain, especially retinal images, the research frontier is shifting from merely achieving transparency to ensuring that this transparency is robust, clinically aligned, and verifiably trustworthy. The next generation of models will need to move beyond single-modality explanations and learn to reason across a combination of fundus images [106], OCT scans, and textual clinical notes to mirror the holistic process of a human expert. Context-aware retinal vision–language models further support this direction by incorporating multiple images, previous diagnoses, and patient-level contextual information, showing the potential of future C-XAI systems to move beyond isolated image explanations toward longitudinal and clinically contextual reasoning [107].

A promising future direction is the development of a higher-level, clinically grounded form of XAI that combines Concept Bottleneck Models (CBMs), Logic Explained Networks (LENs), and neurosymbolic AI. CBMs provide a foundational mechanism for forcing the diagnostic pathway to pass through human-understandable concepts before reaching the final prediction [8]. In retinal diagnosis, this means that visual evidence can first be mapped into clinically meaningful concepts, such as lesions, vessels, macular abnormalities, or optic-disc changes, rather than being represented only through opaque latent features.

However, concept prediction alone is not sufficient for a fully reliable explanation. Logic Explained Networks (LENs) extend concept-based modelling by expressing the relationship between concepts and predictions through First-Order Logic rules [9]. This is important because ophthalmic reasoning often depends on combinations of findings, their presence or absence, and their spatial or clinical relationships. Therefore, LENs can support human-understandable explanations while also enabling the discovery of data-driven logical patterns that may not be visible through simple concept importance scores.

Neurosymbolic AI can further strengthen this direction by connecting learned concepts and logical explanations to formal clinical knowledge. Recent work has highlighted the role of neurosymbolic systems in moving from neural activations toward structured concepts and symbolic reasoning [74]. In retinal C-XAI, this could be achieved by using an ontology-grounded knowledge graph as a semantic layer that defines ophthalmic entities, attributes, and relationships. Such structures can support querying, traceability, and consistency checking, allowing explanations to be evaluated not only for readability but also for logical and clinical soundness.

This direction is already beginning to emerge in ophthalmology. Miladinovic et al. [15] propose a neurosymbolic framework for retinal disease classification from OCT images, showing how learned visual representations can be combined with explicit reasoning constraints. Similarly, ophthalmic knowledge graphs have been explored for structuring clinical records and supporting interpretable decision-making [108]. Ontology-based representations of eye diseases also provide a route for organising clinical concepts in a formal and machine-readable way [109].

Taken together, the integration of CBMs, LENs, and neurosymbolic AI could support a move toward Level 3 XAI. In such a framework, neural encoding captures visual patterns from retinal images, perceptual grounding links those patterns to image-level evidence, conceptual grounding maps them to clinically meaningful biomarkers, LENs express diagnostic reasoning through logical rules, and ontology-based neurosymbolic reasoning validates whether the explanation is clinically coherent. This would allow future retinal AI systems to move beyond isolated concept scores and instead represent explanations as part of a structured clinical reasoning graph [110].

Future work should also validate the generalisability of the proposed C-XAI framework by extending its application to other trustworthiness-critical domains, such as autonomous driving systems, particularly for hazard perception and explainability. One promising direction is to investigate whether C-XAI can systematically audit a model's concept-level reasoning in complex driving scenes and assess whether its internal decision process aligns with verified safety knowledge. For example, integrating structured safety representations such as fault trees could help evaluate whether the model's explanations are not only human-interpretable, but also safety-consistent and causally meaningful.

However, the most pressing challenge remains: how do we trust the explanations of our transparent models? Even a self-explanatory model can be perfectly faithful to its own flawed or incomplete reasoning. Therefore, a promising direction lies in developing rigorous, adversarial audit protocols. This is precisely where SMILE can play an important role as a post-hoc, model-agnostic auditing tool [111]. By using SMILE to independently cross-validate the claims of a self-explainable model, future systems could create an essential "second opinion" that checks for non-clinical shortcuts and builds a deeper, more resilient form of trust. This move from accepting transparency at face value to actively challenging it is the critical next step toward building AI systems that are not just explainable, but truly reliable partners in clinical diagnosis.

5 Conclusion

How can we leverage the remarkable accuracy these systems now offer while still giving clinicians the level of trust, clarity, and accountability they need? This is the challenge of AI's future.

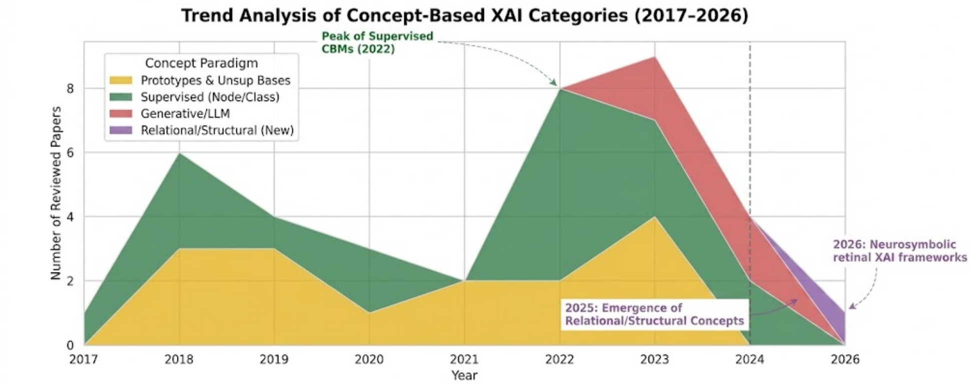

Fig. 15 The evolution of the C-XAI landscape. Supervised methods peaked in 2022, after which the field shifted toward Generative Auditing (2023–2024) and Relational Concept Discovery (2025). Extending this trajectory into 2026, emerging neurosymbolic frameworks further reinforce this structural shift. For example, Miladinović et al. [15] propose a hybrid neural–symbolic framework for OCT-based retinal disease diagnosis, integrating learned representations with explicit reasoning constraints.

This paper explored the fast-growing field of Concept-Based Explainable AI (C-XAI), tracing its development from early feature-attribution approaches to the new directions emerging in 2025, such as Relational Concept Discovery, Mechanistic Interpretability, and Generative Auditing. The unified taxonomy introduced in this study shows how diagnostic AI is beginning to shift. Instead of simply detecting individual features in isolation, newer systems are beginning to capture the relationships and causal structures that matter to how clinicians actually reason about the disease. At the same time, the analysis makes something else very clear: transparency on its own is not enough. Issues like information leakage, hallucinations from generative components, and fragile internal reasoning pathways remind us that a model capable of explaining is not necessarily one we should trust to explain itself correctly. This is why the next step for medical AI has to move beyond explainability toward accountability [112]. Forging this path is the only way to build AI systems that are not merely intelligent tools, but They are truly reliable partners in the high-stakes mission of patient care.

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Consent for publication. Not applicable.

Data availability. Data sharing is not applicable to this article as no new datasets were generated or analysed during the current study.

Materials availability. Not applicable.

Code availability. Not applicable.

Author contribution. Mohadeseh Mollapour: conceptualisation, literature review, methodology, writing—original draft, and editing. Koorosh Aslansefat: supervision, conceptual guidance, and manuscript review. Bhupesh Kumar Mishra: supervision, methodological guidance, and manuscript review. Zhibao Mian: manuscript review and academic feedback.

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